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A Complete Recipe for Diffusion Generative Models

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Abstract

*Score-based Generative Models (SGMs) have demonstrated exceptional synthesis outcomes across various tasks. However, the current design landscape of the forward diffusion process remains largely untapped and often relies on physical heuristics or simplifying assumptions. Utilizing insights from the development of scalable Bayesian posterior samplers, we present a complete recipe for formulating forward processes in SGMs, ensuring convergence to the desired target distribution. Our approach reveals that several existing SGMs can be seen as specific manifestations of our framework. Building upon this method, we introduce Phase Space Langevin Diffusion (PSLD), which relies on score-based modeling within an augmented space enriched by auxiliary variables akin to physical phase space. Empirical results exhibit the superior sample quality and improved speed-quality trade-off of PSLD compared to various competing approaches on established image synthesis benchmarks. Remarkably, PSLD achieves sample quality akin to state-of-the-art SGMs (FID: **2.10** for unconditional CIFAR-10 generation). Lastly, we demonstrate the applicability of PSLD in conditional synthesis using pre-trained score networks, offering an appealing alternative as an SGM backbone for future advancements. Code and model checkpoints can be accessed at* https://github.com/mandt-lab/PSLD.

1. Introduction

Score-based Generative Models [15, 50, 53, 56] are a class of explicit-likelihood based generative models that have recently demonstrated impressive performance on various synthesis benchmarks, such as image generation [7, 16, 39, 42, 43], video synthesis [17, 62] and 3D shape generation [32, 68]. SGMs employ a forward stochastic process to add noise to data incrementally, transforming the data-generating distribution to a tractable prior distribution that enables sampling. Subsequently, a learnable reverse process transforms the prior distribution back to the data distribution using a parametric estimator of the gradient field of the log probability density of the data (a.k.a score).



Figure 1: Unconditional PSLD generated samples. AFHQv2 128 x 128 (Top), CelebA-64 (Bottom Left, FID=2.01) and CIFAR-10 (Bottom Right, FID=2.10)

However, a principled framework for extending the current design space of diffusion processes is still missing. Although some studies have proposed augmenting the forward diffusion process with auxiliary variables [9] to improve sample quality, their design is primarily motivated by physical intuition and non-obvious how to generalize. Therefore, a principled framework is required to explore the space of possible diffusion processes better.

In this work, we propose a *complete* recipe for the design of diffusion processes, motivated by the design of stochastic gradient MCMC samplers [5, 33, 61]. Our recipe leads to a flexible parameterization of the forward diffusion process without requiring physical intuition. Moreover, under the proposed parameterization, the forward process is guaranteed to converge to a prior distribution of interest. We show that several existing SGMs can be cast under our diffusion process parameterization. Furthermore, using our proposed recipe, we introduce PSLD, a novel SGM which performs diffusion in the joint space of data and auxiliary variables. We demonstrate that PSLD generalizes Critically Damped Langevin Diffusion (CLD) [9] and outperforms existing baselines on several empirical settings on standard image synthesis benchmarks such as CIFAR-10 [25] and CelebA-64 [30]. More specifically, we make the following

theoretical and empirical contributions:

1. **A Complete Recipe for SGM Design:** We propose a specific parameterization of the forward process, guaranteed to converge asymptotically to a desired stationary "prior" distribution. The proposed recipe is *complete* in the sense that it subsumes all possible Markovian stochastic processes which converge to this distribution. We show that several existing SGMs [9, 56] can be cast as specific instantiations of our recipe.

2. **Phase Space Langevin Diffusion(PSLD):** To exemplify the proposed diffusion parameterization concretely, we propose PSLD: a novel SGM which performs diffusion in the phase space by adding noise in **both** data and the momentum space.

3. **Superior Sample Quality and Speed-Quality Trade-offs:** Using ablation experiments on standard image synthesis benchmarks like CIFAR-10 and CelebA-64, we demonstrate the benefits of adding stochastic noise in both the data and the momentum space on overall sample quality and the speed quality trade-offs associated with PSLD. Furthermore, using similar score network architectures, our proposed method outperforms existing diffusion baselines on both criteria across different sampler settings.

4. **State-of-the-Art Sample Quality:** We show that PSLD outperforms competing baselines and achieves competitive perceptual sample quality to other state-of-the-art methods. Our model achieves an FID [13] score of **2.10**, an IS score [45] of **9.93** on unconditional CIFAR-10 and an FID score of **2.01** on CelebA-64.

5. **Conditional synthesis:** We show that pre-trained unconditional PSLD models can be used for conditional synthesis tasks like class-conditional generation and image inpainting during inference.

Overall, given the superior performance of PSLD on several tasks, we present an attractive alternative to existing SGM backbones for further development. We organize the rest of our work as follows: Section 2 presents some background on SGMs and our proposed *recipe* for SGM design. Section 3 presents the construction of our novel PSLD model. Section 4 presents our empirical findings. Lastly, Section 5 compares our proposed contributions to several existing works while we present some directions for future work in Section 6.

2. A Complete Recipe for SGM Design

2.1. Background

Consider the following forward process SDE for converting data $\mathbf{x}_t \in \mathbb{R}^d$ to noise,

$$d\mathbf{x}_t = \boldsymbol{f}(\mathbf{x}_t, t)\, dt + \boldsymbol{G}(t)\, d\mathbf{w}_t, \quad t \in [0, T],$$

with continuous time variable $t \in [0, T]$, a standard Wiener process $\mathbf{w}_t$, drift coefficient $\boldsymbol{f} \colon \mathbb{R}^d \times [0, T] \to \mathbb{R}^d$, and diffusion coefficient $\boldsymbol{G} \colon [0, T] \to \mathbb{R}^{d \times d}$. Given this forward process, the corresponding reverse-time diffusion process [1, 56] that generates data from noise is given by

$$d\mathbf{x}_t = \bar{\boldsymbol{f}}(\mathbf{x}_t, t)\, dt + \boldsymbol{G}(t) d\bar{\mathbf{w}}_t, \tag{1}$$

$$\bar{\boldsymbol{f}}(\mathbf{x}_t, t) = \left[\boldsymbol{f}(\mathbf{x}_t, t) - \boldsymbol{G}(t)\boldsymbol{G}(t)^\top \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)\right].$$

Given an estimate of the *score* $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ of the marginal distribution over $\mathbf{x}_t$ at time t, the reverse SDE can then be simulated to recover the original data samples from noise. In practice, the score is intractable to compute and is approximated using a parametric estimator $s_\theta(\mathbf{x}_t, t)$, trained using denoising score matching [15, 53, 56, 60]:

$$\min_{\boldsymbol{\theta}} \mathbb{E}_t \mathbb{E}_{p(\mathbf{x}_0)} \mathbb{E}_{p_t(\mathbf{x}_t|\mathbf{x}_0)} \left[\lambda(t) \|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{x}_0)\|_2^2\right].$$

Above, the time t is usually sampled from a uniform distribution $\mathcal{U}(0, T)$. Given an appropriate choice of $\boldsymbol{f}$ and $\boldsymbol{G}$, the perturbation kernel $p(\mathbf{x}_t|\mathbf{x}_0)$ can frequently be computed analytically (e.g., it is typically Gaussian). Consequently, samples $\mathbf{x}_t$ can be generated in constant time, allowing for fast stochastic gradient updates. The choice of the weighting schedule $\lambda(t)$ plays an essential role during training and can be selected to optimize for likelihood [52] or sample quality [56]. The forward SDE asymptotically converges to an equilibrium distribution (usually a standard isotropic Gaussian) which can be used as a prior to initialize the reverse SDE, which can then be simulated using numerical solvers.

2.2. A General Recipe for Constructing Stochastic Forward Processes

As has been shown in the MCMC literature [3, 5], it is often beneficial to extend the sampling space into an *augmented* space according to $\mathbf{z} = [\mathbf{x}, \mathbf{m}]^T \in \mathbb{R}^{d_z}$, where $\mathbf{x} \in \mathbb{R}^{d_x}$ is the original state space variable and $\mathbf{m} \in \mathbb{R}^{d_m}$ corresponds to some additional auxiliary dimensions. Simulating the dynamics of the variable $\mathbf{z}$ may have desirable properties, such as faster mixing. Inspired by the naming conventions in statistical physics, we call $\mathbf{x}$ the *position* variable and $\mathbf{m}$ the *momentum* variable. Accordingly, we denote their joint space as *augmented space* (or *phase space if* $\mathbf{x}$ *and* $\mathbf{m}$ have equal dimensions). Note that our notation also captures the scenario where $\mathbf{m}$ is absent (zero-dimensional). We now consider the following form of the stochastic process modeled using an Itô SDE:

$$d\mathbf{z} = \boldsymbol{f}(\mathbf{z})dt + \sqrt{2\boldsymbol{D}(\mathbf{z})}d\mathbf{w}_t, \tag{2}$$

with drift term $\boldsymbol{f}(\mathbf{z}) \in \mathbb{R}^{d_z}$ and diffusion coefficient $\boldsymbol{D}(\mathbf{z}) \in \mathbb{R}^{d_z \times d_z}$. We assume a desired stationary state distribution $p_s(\mathbf{z})$ specified as

$$p_s(\mathbf{z}) \propto \exp(-H(\mathbf{z})),$$

$$H(\mathbf{z}) = H(\mathbf{x}, \mathbf{m}) = U(\mathbf{x}) + \frac{\mathbf{m}^T M^{-1} \mathbf{m}}{2}, \tag{3}$$

where H represents the Hamiltonian associated with $p_s(\mathbf{z})$. The first term in $H(\mathbf{z})$ represents the potential energy $U(\mathbf{x})$ associated with the configuration $\mathbf{x}$ while the second term represents the kinetic energy associated with the auxiliary (or momentum) variables $\mathbf{m}$ and mass matrix $M\boldsymbol{I}_{d_m}$. In the context of Bayesian inference, [33] propose a framework to elucidate the design space of possible MCMC samplers that sample from $p_s(\mathbf{z})$. In this framework, the drift $\boldsymbol{f}(\mathbf{z})$ can be parameterized as

$$\boldsymbol{f}(\mathbf{z}) = -(\boldsymbol{D}(\mathbf{z}) + \boldsymbol{Q}(\mathbf{z}))\nabla H + \tau(\mathbf{z}), \tag{4}$$

$$\tau_i(\mathbf{z}) = \sum_{j=1}^{d} \frac{\partial}{\partial \mathbf{z}_j} \left(\boldsymbol{D}_{ij}(\mathbf{z}) + \boldsymbol{Q}_{ij}(\mathbf{z})\right),$$

where $\boldsymbol{Q}(\mathbf{z})$ represents a skew-symmetric curl matrix. Furthermore, the following result holds:

Theorem 2.1 (Yin et. al. [64])**.** *For the dynamics defined in Eqn. 2, if $\boldsymbol{f}(\mathbf{z})$ is parameterized as in Eqn. 4 with $\boldsymbol{D}(\mathbf{z})$ positive semidefinite and $\boldsymbol{Q}(\mathbf{z})$ skew-symmetric, then the distribution $p_s(\mathbf{z}) \propto \exp(-H(\mathbf{z}))$ is a stationary distribution for the dynamics.*

Theorem 2.1 implies that for a specific choice of matrices $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$, the process defined in Eqn. 2 always asymptotically samples from the target distribution $p_s(\mathbf{z})$. Moreover, [33] showed in the context of MCMC that the parameterization defined in Eqn. 4 is *complete* as follows:

Theorem 2.2 (Ma et. al. [33])**.** *Assume the stochastic process in Eqn. 2 converges to a unique stationary distribution $p_s(\mathbf{z})$. Then, under mild regularity assumptions, there exists a corresponding skew-symmetric matrix $\boldsymbol{Q}(\mathbf{z})$, such that $\boldsymbol{f}(\mathbf{z})$ assumes the form of Eqn. 4.*

We include the proofs for Theorems 2.1 and 2.2 in Appendix A.1 for completeness. These results provide a general recipe for designing forward processes in SGMs.

For the SGM to be a useful forward process, we need it to converge to a simple factorized distribution that serves as the initialization point of the backwards (generative) process. Consequently, we consider the following form of the stationary distribution $p_s(\mathbf{z})$:

$$p_s(\mathbf{z}) = \mathcal{N}(\mathbf{x}; \mathbf{0}_{d_x}, \boldsymbol{I}_{d_x})\mathcal{N}(\mathbf{0}_{d_m}, M\boldsymbol{I}_{d_m}). \tag{5}$$

This form results from setting $U(\mathbf{x}) = \frac{\mathbf{x}^T\mathbf{x}}{2}$ in Eqn. 3. Therefore, for a positive semidefinite matrix $\boldsymbol{D}(\mathbf{z})$ and a skew-symmetric matrix $\boldsymbol{Q}(\mathbf{z})$, the most general class of forward processes which lead to an invariant distribution $p_s(\mathbf{z})$ can be specified by substituting the form of $\nabla H(\mathbf{z})$ (corresponding to $p_s(\mathbf{z})$ defined in Eqn. 5) in Eqn. 4. A similar characterization of forward processes has also been explored in a concurrent work by [48] in the context of likelihood estimation; see Section 5 for more details.

2.3. Additional constraints on $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$

Theorems 2.1 and 2.2 show that the proposed forward process parameterization is complete upon specifying the target distribution $p_s(\mathbf{z})$ (such as Eqn. 5). However, we need additional requirements for the resulting generative model and the corresponding training objective to be tractable. Specifically, when using the denoting score matching objective [60], we require the perturbation kernel $p(\mathbf{z}_t|\mathbf{z}_0)$ to be computable in closed form. In practice, this restricts our possible choices for $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$ to constant matrices (i.e., independent of the state variable $\mathbf{z}$). Yet, even with this requirement, the framework provides a large design space of models. We provide several examples of existing SGMs that can be understood as special cases of our recipe in Appendix A.2. We stress that training paradigms other than denoting score matching (e.g., such as Sliced Score matching [55]) may enable a wider range of possible models with non-constant matrices D and Q.

3. Phase Space Langevin Diffusion

We next use the proposed recipe to construct a specific SGM with favorable properties.

3.1. Model Definition

We restrict the family of forward processes considered in this work by constraining $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$ as constant matrices, i.e., independent of state $\mathbf{z}$. Moreover, we assume that $\mathbf{x}$ and $\mathbf{m}$ have the same dimension d, i.e. $\mathbf{z} \in \mathbb{R}^{2d}$. Consequently, the drift $f(\mathbf{z})$ becomes affine in $\mathbf{z}$ and the perturbation kernel $p(\mathbf{z}_t|\mathbf{z}_0)$ can be computed analytically [47]. Among the possible samplers, we choose a specific form involving $d-$dimensional position and momentum coordinates, $\mathbf{z}_t = [\mathbf{x}_t, \mathbf{m}_t]^T$ where $\mathbf{x}_t \in \mathbb{R}^d$, $\mathbf{m}_t \in \mathbb{R}^d$. Our choice for $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$ is as follows:

$$\boldsymbol{D} \coloneqq \frac{\beta}{2}\left(\begin{pmatrix}\Gamma & 0\\ 0 & M\nu\end{pmatrix} \otimes \boldsymbol{I}_d\right), \quad \boldsymbol{Q} \coloneqq \frac{\beta}{2}\left(\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix} \otimes \boldsymbol{I}_d\right). \tag{6}$$

Above, Γ, M, ν and β are positive scalars. Along with these choices of $\boldsymbol{D}$ and $\boldsymbol{Q}$, we have $\tau(\mathbf{z}) = \mathbf{0}$. The resulting forward process is given by:

$$d\mathbf{z}_t = \boldsymbol{f}(\mathbf{z}_t)dt + \boldsymbol{G}(t)d\mathbf{w}_t, \tag{7}$$

$$\boldsymbol{f}(\mathbf{z}_t) = \left(\frac{\beta}{2}\begin{pmatrix}-\Gamma & M^{-1}\\ -1 & -\nu\end{pmatrix} \otimes \boldsymbol{I}_d\right)\mathbf{z}_t,$$

$$\boldsymbol{G}(t) = \sqrt{2D(\mathbf{z}_t)} = \begin{pmatrix}\sqrt{\Gamma\beta} & 0\\ 0 & \sqrt{M\nu\beta}\end{pmatrix} \otimes \boldsymbol{I}_d.$$

We denote the form of the SDE in Eqn. 7 as the *Phase Space Langevin Diffusion (PSLD)*. Note that PSLD generalizes Critically Damped Langevin Diffusion (CLD) proposed in [9], which can be obtained by setting $\Gamma = 0$, $\bar{\nu} = M\nu$, and $\bar{\beta} = \frac{\beta}{2}$. Like CLD, the parameter M^{-1} couples the data space state $\mathbf{x}_t$ with the auxiliary state $\mathbf{m}_t$. The parameters β, Γ, and ν control the amount of noise in the forward SDE. Without loss of generality, we use a time-independent β. However, unlike CLD or any physical system, PSLD adds stochastic noise in the data space *in addition* to the noise injected into the momentum component of phase space. While we are not aware of any physical system that displays such behavior, it is a valid stochastic process compatible with our framework. Our experiments reveal the strong benefits of having these two independent noise sources.

Furthermore, CLD [9] proposes setting $\bar{\nu}^2 = 4M$, corresponding to critical damping in a physical system. Under critical damping, an ideal balance is achieved between the oscillatory Hamiltonian dynamics and the noise-injecting Ohrnstein-Uhlenbeck (OU) term, leading to faster convergence to equilibrium. We generalize this line of argument in Appendix B.1, where we derive $(\Gamma - \nu)^2 = 4M^{-1}$ as the equivalent condition for critical damping in PSLD. Throughout this work, we choose Γ, ν, and M^{-1} such that the critical damping condition in PSLD is satisfied.

3.2. PSLD Training

Since the drift coefficient in PSLD is affine, the perturbation kernel $p(\mathbf{z}_t|\mathbf{z}_0)$ of PSLD can be computed analytically. We can then use DSM to learn the score function $\boldsymbol{s}_\theta(\mathbf{z}_t, t)$. More specifically, following the derivation in [52], it can be shown that the Maximum-Likelihood (ML) based DSM objective for PSLD can be specified as (Proof in Appendix B.2.1)

$$\min_{\boldsymbol{\theta}} \mathbb{E}_t \mathbb{E}_{p(\mathbf{z}_0)} \mathbb{E}_{p_t(\mathbf{z}_t|\mathbf{z}_0)} \Big[\mathcal{L}_x(\boldsymbol{\theta}, \mathbf{z}_t, \mathbf{z}_0) + \mathcal{L}_m(\boldsymbol{\theta}, \mathbf{z}_t, \mathbf{z}_0) \Big], \tag{8}$$

$$\mathcal{L}_x = \Gamma\beta \| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{z}_t, t)|_{0:d} - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{z}_t|\mathbf{z}_0) \|_2^2,$$
$$\mathcal{L}_m = M\nu\beta \| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{z}_t, t)|_{d:2d} - \nabla_{\mathbf{m}_t} \log p_t(\mathbf{z}_t|\mathbf{z}_0) \|_2^2,$$

where $\boldsymbol{s}_{\boldsymbol{\theta}}(\mathbf{z}_t, t)|_{0:d}$ and $\boldsymbol{s}_{\boldsymbol{\theta}}(\mathbf{z}_t, t)|_{d:2d}$ represent the first and the last d components of the vector $\boldsymbol{s}_\theta(\mathbf{z}_t, t)$ respectively. In the above DSM objective, the perturbation kernel $p(\mathbf{z}_t|\mathbf{z}_0) = \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ is a multivariate Gaussian while $p(\mathbf{z}_0) = p(\mathbf{x}_0)\mathcal{N}(\mathbf{m}_0; 0, M\gamma\boldsymbol{I}_d)$, where $p(\mathbf{x}_0)$ is the data distribution. In this work, we reformulate the DSM objective in Eqn. 8 as follows (also see Appendix B.2.1):

$$\min_{\boldsymbol{\theta}} \mathbb{E}_t \mathbb{E}_{p(\mathbf{z}_0)} \mathbb{E}_{p_t(\mathbf{z}_t|\mathbf{z}_0)} \Big[\lambda(t) \| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{z}_t, t) - \nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t|\mathbf{z}_0) \|_2^2 \Big].$$

Furthermore, due to its gradient variance reduction properties, we instead use the Hybrid Score Matching (HSM) objective [9] by marginalizing out the momentum variables $\mathbf{m}_0$ as $p(\mathbf{z}_t|\mathbf{x}_0) = \int p(\mathbf{z}_t|\mathbf{x}_0, \mathbf{m}_0) p(\mathbf{m}_0) d\mathbf{m}_0$. Since both distributions $p(\mathbf{z}_t|\mathbf{x}_0, \mathbf{m}_0)$ and $p(\mathbf{m}_0)$ are Gaussian, $p(\mathbf{z}_t|\mathbf{x}_0)$ will also be a Gaussian.

Score Network Parameterization: Since the perturbation kernel $p(\mathbf{z}_t|\mathbf{x}_0)$ in the HSM objective is also a multivariate Gaussian, we have $p(\mathbf{z}_t|\mathbf{x}_0) \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$. Furthermore, let $\boldsymbol{\Sigma}_t = \boldsymbol{L}_t \boldsymbol{L}_t^T$ be the Cholesky factorization of the matrix $\boldsymbol{\Sigma}_t$. We have

$$\nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t|\mathbf{x}_0) = -\boldsymbol{\Sigma}_t^{-1}(\mathbf{z}_t - \boldsymbol{\mu}_t) = -\boldsymbol{L}_t^{-T}\boldsymbol{\epsilon}, \tag{9}$$

where $\boldsymbol{L}_t^{-T}$ is the transposed inverse of the $\boldsymbol{L}_t$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0_{2d}}, \mathbf{I_{2d}})$. Therefore, we parameterize our score function estimator as $\boldsymbol{s}_\theta(\boldsymbol{z}_t, t) = -\boldsymbol{L}_t^{-T}\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)$. Although alternative parameterizations of the score network $s_\theta(\mathbf{z}_t, t)$ like *mixed score* can be possible [9, 20, 58], we do **not** explore such parameterizations in this work and leave further exploration to future work. We provide additional details on the score network parameterization in PSLD in Appendix B.2.2 and the analytical form of the perturbation kernel $p(\mathbf{z}_t|\mathbf{x}_0)$ in Appendix B.3.

Final Training Objective: Using our score parameterization from Eqn. 9 with $\lambda(t) = \frac{1}{\|\boldsymbol{L}_t^{-T}\|_2^2}$, we get the following *epsilon-prediction* form of the HSM objective (See Appendix B.2.3 for a complete derivation):

$$\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim\mathcal{U}(0,T)} \mathbb{E}_{p(\mathbf{x}_0)} \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\boldsymbol{I}_d)} \left[\|\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{\mu}_t + \boldsymbol{L}_t\boldsymbol{\epsilon}, t) - \boldsymbol{\epsilon}\|_2^2 \right]. \tag{10}$$

The epsilon-prediction objective has been shown to generate superior sample quality [9, 15, 56]. In this work, we optimize for sample quality and therefore use this objective for training all models. One key difference between the objective in Eqn. 10 and the HSM objective in CLD is that, unlike CLD, we predict the full 2d-dimensional $\boldsymbol{\epsilon}$ due to the structure of our diffusion coefficient $\boldsymbol{G}(t)$ (see Appendix B.2 for more details). Therefore, for a non-zero Γ, the neural-net-based score predictor in PSLD has twice the number of output channels as in CLD. However, the increase in parameters due to this architectural update is negligible.

3.3. PSLD Sampling

Following the result from [56], the reverse process SDE corresponding to the forward process SDE defined in Eqn. 8 can be formulated as follows:

$$d\bar{\mathbf{z}}_t = \bar{\boldsymbol{f}}(\bar{\mathbf{z}}_t)dt + \boldsymbol{G}(T-t)d\bar{\mathbf{w}}_t, \tag{11}$$

$$\bar{\boldsymbol{f}}(\bar{\mathbf{z}}_t) = \frac{\beta}{2} \begin{pmatrix} \Gamma\bar{\mathbf{x}}_t - M^{-1}\bar{\mathbf{m}}_t + 2\Gamma\boldsymbol{s}_\theta(\bar{\mathbf{z}}_t, T-t)|_{0:d} \\ \bar{\mathbf{x}}_t + \nu\bar{\mathbf{m}}_t + 2M\nu\boldsymbol{s}_\theta(\bar{\mathbf{z}}_t, T-t)|_{d:2d} \end{pmatrix},$$

$$\boldsymbol{G}(T-t) = \begin{pmatrix} \sqrt{\Gamma\beta} & 0 \\ 0 & \sqrt{M\nu\beta} \end{pmatrix} \otimes \boldsymbol{I}_d,$$

Method	Size	NFE	FID@50k (↓)
Ours (Baseline)			
CLD (w/o MS)	97M	1000	2.41
Ours (Proposed)			
PSLD (Γ=0.02)	39M	1000	2.80
PSLD (Γ=0.01)	55M	1000	2.34
PSLD (Γ=0.02)	55M	1000	2.30
PSLD (Γ=0.01)	97M	1000	2.23
PSLD (Γ=0.02)	97M	1000	**2.21**
CLD (w/ MS) [9]	108M	1000	2.27
VPSDE (deep)† [56]	108M	1000	2.46
VESDE (deep)† [56]	108M	1000	2.43
DDPM [15]	35.7M	1000	3.17
iDDPM [36]	-	1000	2.90
DiffuseVAE [38]	35.7M	1000	2.80
NCSNv2 [54]	-	-	10.87
NCSN [53]	-	1000	25.32
VDM [21]	-	1000	7.41

Table 1: PSLD (SDE) sample quality comparisons for CIFAR-10. PSLD outperforms competing SDE baselines for a similar sampling budget. FID computed using 50k samples. MS: Mixed Score †: Results from [9].

Method	Size	NFE	FID@50k (↓)
Ours (Baseline)			
CLD (w/o MS)	97M	352	2.80
Ours (Proposed)			
PSLD (Γ=0.01)	55M	243	2.41
PSLD (Γ=0.02)	55M	232	2.40
PSLD (Γ=0.01)	97M	246	**2.10**
PSLD (Γ=0.02)	97M	231	2.31
Ours (Proposed)			
PSLD (Γ=0.01)	97M	159	2.13
PSLD (Γ=0.02)	97M	159	2.34
LSGM† [58]	100M	131	4.60
LSGM [58]	476M	138	**2.10**
VPSDE† [56]	108M	141	2.76
CLD (w/ MS) [9]	108M	147	2.71
CLD (w/ MS) [9]	108M	312	2.25
ScoreFlow (VP) [52]	108M	-	5.34
Flow Matching (w/ OT) [28]	-	142	6.35
DDIM (VPSDE)† [51]	108M	150	3.15

Table 2: PSLD (ODE) sample quality comparisons for CIFAR-10. PSLD outperforms most competing ODE baselines. FID computed using 50k samples. MS: Mixed Score. †: Results from [9].

where $\bar{\mathbf{z}}_t = \mathbf{z}_{T-t}$, $\bar{\mathbf{x}}_t = \mathbf{x}_{T-t}$, $\bar{\mathbf{m}}_t = \mathbf{m}_{T-t}$. We can simulate this reverse process SDE using standard numerical SDE solvers like the Euler-Maruyama (EM) sampler [23]. As an alternative, [9] propose SSCS: a symmetric splitting-based integrator and show that SSCS exhibits a better speed-sample quality tradeoff than EM. Consequently, we extend SSCS for PSLD by using the following splitting formulation:

$$\begin{pmatrix} d\bar{\mathbf{x}}_t \\ d\bar{\mathbf{m}}_t \end{pmatrix} = \mathcal{A} + \mathcal{S}, \tag{12}$$

$$\mathcal{A} = \frac{\beta}{2} \begin{pmatrix} -\Gamma\bar{\mathbf{x}}_t - M^{-1}\bar{\mathbf{m}}_t \\ \bar{\mathbf{x}}_t - \nu\bar{\mathbf{m}}_t \end{pmatrix} dt + \boldsymbol{G}(T-t)d\bar{\mathbf{w}}_t,$$

$$\mathcal{S} = \beta \begin{pmatrix} \Gamma\bar{\mathbf{x}}_t + \Gamma\boldsymbol{s}_\theta(\bar{\mathbf{z}}_t, T-t)|_{0:d} \\ \nu\bar{\mathbf{m}}_t + M\nu\boldsymbol{s}_\theta(\bar{\mathbf{z}}_t, T-t)|_{d:2d} \end{pmatrix} dt.$$

Indeed for $\Gamma = 0$, the sampler in Eqn. 12 resembles the SSCS sampler proposed in [9]. It is worth noting that despite an updated formulation, the order of the SSCS sampler, as analyzed in [9], remains unchanged. We discuss the exact solution of the analytical part of the Modified-SSCS sampler in Eqn. 12 and other relevant details in Appendix B.4.2.

4. Experiments

Datasets: We run experiments on three datasets: CIFAR-10 [25], CelebA [30] at 64 x 64 resolution and the AFHQv2 [6] dataset at 128 x 128 resolution.

Baselines: We primarily compare PSLD with two popular SGM baselines: VP-SDE [56] and CLD [9] (a particular case of PSLD with $\Gamma = 0$). For PSLD and CLD, unless specified otherwise, we operate in the critical damping regime with a fixed $M^{-1} = 4$ and therefore choose Γ and ν accordingly ($\nu = 2\sqrt{M^{-1}} + \Gamma, \nu \geq 0, \Gamma \geq 0$).

Metrics: We use the FID [13] score for quantitatively assessing sample quality, while we use NFE (Number of Function Evaluations) to assess the sampling efficiency of all methods.

We provide full implementation details in Appendix C. The rest of our experimental section is organized as follows: Firstly, we compare the state-of-the-art performance of PSLD with popular SGM baselines on unconditional image generation. We show that PSLD outperforms competing baselines for similar compute budgets. Secondly, as an ablation experiment, we empirically and theoretically analyze the impact of the SDE parameters Γ and ν on downstream sample quality in PSLD. Furthermore, we analyze the speed-quality trade-off in PSLD and show that PSLD yields better sample quality than competing baselines across four different sampler settings. Lastly, we show that pre-trained unconditional PSLD models can be used for downstream tasks like class-conditional image synthesis and image inpainting.

4.1. State-of-the-art Comparisons

Setup: We now compare the sample quality of our proposed method with existing popular SGM methods on the CIFAR-10 and CelebA-64 datasets for unconditional image synthesis. We use PSLD with $\Gamma \in \{0.01, 0.02\}$ for CIFAR-10 and PSLD with $\Gamma = 0.005$ for CelebA-64 for state-of-the-art (SOTA) comparisons (See Section 4.2 for a theoretical and empirical justification of these choices). Moreover, we use the training objective in Eqn. 10 **without** any alternative score parameterizations (like *mixed score*[9, 58]) to train our models for SOTA comparisons.

Method	NFE	FID@50k
(**Ours**) PSLD ($\Gamma = 0.005$)	250	**2.01**
(**Ours**) PSLD ($\Gamma = 0.005$, ODE)	244	2.56
PNDM [29]	250	2.71
DDIM [51]	250	4.44
VPSDE [56]	1000	2.32
Gamma DDPM [35]	1000	2.92
DDPM [15]	1000	3.26
DiffuseVAE [38]	1000	3.97
VESDE [56]	2000	3.95
NCSN (w/ denoising) [53]	-	25.3
NCSNv2 (w/ denoising) [54]	-	10.23

Table 3: PSLD sample quality comparisons for CelebA-64. FID computed using 50k samples.

	CIFAR-10 (39M)		CelebA-64 (66M)	
Γ	FID@50k ↓ (EM-QS)	FID@50k ↓ (EM-US)	FID@10k ↓ (EM-QS)	FID@10k ↓ (EM-US)
0	3.64	3.60	4.59	4.60
0.005	3.42	3.34	**4.17**	4.37
0.01	3.15	2.94	4.22	4.34
0.02	3.26	**2.80**	4.43	4.52
0.25	4.99	9.48	93.99	95.13

Table 4: Impact of increasing Γ (with fixed M^{-1}) on sample quality (NFE=1000). FID computed using 50k and 10k samples for the CIFAR-10 and CelebA-64 datasets, respectively. QS: Quadratic Striding, US: Uniform Striding.

Unless specified otherwise, we perform sampling using the EM sampler with Uniform Striding (US) for CIFAR-10 and Quadratic Striding (QS) for CelebA-64 for the SDE setup and report FID scores on 50k samples (denoted as FID@50k). We include full details on our SDE and ODE solver setup for SOTA analysis in Appendix C.5. We report the FID scores for most competing methods for a maximum sampling budget of N=1000 while reporting model sizes whenever available for CIFAR-10 for fair comparisons.

Main Observations: Table 1 compares CIFAR-10 sample quality between different methods using stochastic sampling. Our proposed method with $\Gamma = 0.02$ and 39M parameters achieves an FID score of 2.80, outperforming the DDPM [15] baseline while performing comparably with DiffuseVAE [38], for similar model sizes. It is worth noting that DiffuseVAE refines samples generated from a VAE [22] using a DDPM backbone and is complementary to our work. *Furthermore, our larger PSLD model achieves an FID of **2.21**, which is better than CLD [9] (with or without the Mixed Score (MS) parameterization) and (VP/VE)-SDE baselines for similar NFE budget and model sizes*. For CelebA-64 (Table 3), PSLD outperforms the VP/VE-SDE baselines by a significant margin while requiring only 250 NFEs.

We next analyze ODE sample quality in PSLD. Table 2 compares CIFAR-10 sample quality between different methods using ODE-based samplers. PSLD with $\Gamma = 0.01$ achieves an FID score of 2.10 and outperforms most competing methods except LSGM [58]. Though the original LSGM model is more than four times the size of our SOTA model, PSLD performs comparably with LSGM. When scaled to a similar size, LSGM performs much worse than PSLD (FID: 4.60 for LSGM-100M vs. 2.10 for PSLD). We note that EDM [20] achieves an FID of 2.05 on unconditional CIFAR-10 generation (without data augmentation) by analyzing several design choices associated with diffusion models (like score network architectures, loss preconditioning, and sampler design). We did not explore this line of research but note that their approach complements our proposed method, and exploring some of these design choices in the context of PSLD can be an exciting future direction.

Interestingly, PSLD with the ODE setup obtains a better FID score than the SDE setup (FID: 2.21 for SDE vs. 2.10 for ODE) while requiring around four times lesser NFEs. Moreover, when using a solver tolerance of $1e^{-4}$, PSLD achieves an FID score of 2.13, comparable to the best FID of 2.10 while reducing NFEs significantly. *This tradeoff is worse for other SGMs like CLD and VP-SDE (Table 2)*. We report additional SOTA results in Appendix D.3.

4.2. Impact of Γ and ν on PSLD Sample Quality



Figure 2: Impact of increasing $\Gamma = \{0, 0.005, 0.02, 0.25\}$ (Top to Bottom) on CelebA-64 sample quality. The best sample quality is achieved at $\Gamma = 0.005$ (Second Row) while increasing Γ to 0.25 results in loss of high-frequency image features.

Setup and Baselines: Since adding stochastic noise in both the data and the momentum space is one of the primary aspects of PSLD, we now analyze the impact of the choice of Γ and ν on downstream sample quality. For subsequent experimental results, we use our smaller ablation models (for PSLD and relevant baselines) for comparisons. Table 4 shows the impact of varying Γ on sample quality for the CIFAR-10 and CelebA-64 datasets. Our ablation CLD baseline (PSLD with $\Gamma = 0, \nu = 4$) achieves an FID of 3.60

using the Euler-Maruyama (EM) sampler with Uniform striding (US) and 3.64 using the EM sampler with Quadratic striding (QS). Our results are comparable with the FID of 3.56 obtained by [9] for their CLD ablation model on CIFAR-10 without using the mixed-score parameterization. Our VP-SDE ablation baseline (not shown in Table 4) obtains an FID of 3.19 using EM-US (with 1000 NFEs).

Main Observations: We observe that *setting* Γ *to a non-zero value within a specific range improves sample quality significantly over CLD*. Specifically, our ablation CIFAR-10 model achieves FID scores of 2.94 and 2.80 for $\Gamma = 0.01$ and $\Gamma = 0.02$ respectively (with EM-US) and outperforms our VP-SDE and CLD baselines without using alternative score-network parameterizations like mixed-score which is crucial for competitive performance of CLD [9]. We make a similar observation for the CelebA-64 dataset on which our model achieves the best FID of 4.17 using EM-QS and outperforms our CLD baseline (FID: 4.59). Interestingly, the sample quality worsens for both datasets on increasing Γ outside a range. For instance, for CIFAR-10, further increasing Γ from 0.02 to 0.04 (not shown in Table 4) resulted in an increase in FID from 2.80 to 2.95. Consequently, the sample quality for both datasets is the worst at $\Gamma = 4.25, \nu = 0.25$. We also note that EM-US works better than EM-QS for CIFAR-10 and vice-versa for CelebA-64.

Figure 2 further validates our findings qualitatively for the CelebA-64 dataset where for $\Gamma = 0.25$, the score network can only recover high-level semantic structures (like gender and glasses, among others) but is unable to recover high-frequency details. Since the diffusion denoiser recovers most high-frequency information in the low-timestep regime, these observations suggest denoising issues near low-timestep indices. We next provide a formal justification for this observation.

Theoretical justification of adding stochasticity in the position space: Since PSLD involves adding stochasticity in both the data and the momentum space, during training, we need to predict the noise $\boldsymbol{\epsilon}_\theta^x(\mathbf{z}_t, t)$ and $\boldsymbol{\epsilon}_\theta^m(\mathbf{z}_t, t)$ in both the data and the momentum space respectively. Therefore, it is unclear why PSLD leads to better sample quality than CLD since predicting both noise components can lead to additional sources of errors during sampling.

However, in the context of the EM sampler, we find (see Appendix D.1) that setting a *small* non-zero Γ can significantly suppress prediction errors from $\boldsymbol{\epsilon}_\theta^x(\mathbf{z}_t, t)$ at the expense of introducing negligible extra errors from $\boldsymbol{\epsilon}_\theta^m(\mathbf{z}_t, t)$. Contrarily, using larger values of Γ results in scaling the prediction errors from $\boldsymbol{\epsilon}_\theta^x(\mathbf{z}_t, t)$ by a significant factor, especially in the low-timestep regime, leading to worse sample quality with significant degradations in high-frequency sample details as observed in Figure 2.

		NFE (FID@10k ↓)				
Sampler	Method	50	100	250	500	1000
EM-QS	CLD	25.01	8.91	5.97	5.61	5.7
	VP-SDE	**17.72**	7.45	5.59	5.51	5.51
	(Ours) PSLD	19.94	**7.33**	**5.26**	**5.20**	**5.28**
EM-US	CLD	119.68	45.60	**9.08**	5.71	5.65
	VP-SDE	**84.54**	41.93	12.61	5.92	5.19
	(Ours) PSLD	100.62	**39.96**	11.26	**5.45**	**4.82**
SSCS-QS	CLD	21.31	8.37	5.82	5.75	5.69
	(Ours) PSLD	**16.12**	**7.16**	**5.36**	**5.35**	**5.27**
SSCS-US	CLD	75.45	24.74	6.09	5.74	5.78
	(Ours) PSLD	**72.42**	**20.46**	**5.19**	**4.92**	**5.29**

Table 5: PSLD exhibits better speed vs. sample quality tradeoffs over competing baseline SDEs (CLD and VP-SDE) on CIFAR-10 across four samplers configurations. The rightmost five columns indicate NFEs, with **bold** indicating the best result for that sampler. QS: Quadratic Striding, US: Uniform Striding. See Appendix D.2 for extended results.

Therefore, intuitively, Γ *introduces a trade-off between error contribution from both noise predictors* $\boldsymbol{\epsilon}_\theta^x(\mathbf{z}_t, t)$ *and* $\boldsymbol{\epsilon}_\theta^m(\mathbf{z}_t, t)$ *with small values of* Γ *providing a favorable trade-off which improve overall sample quality*. As a general guideline, we find $\Gamma = 0.01$ to work well across datasets. Figure 1 shows some qualitative samples generated from PSLD trained on the AFHQv2 [6] dataset with $\Gamma = 0.01, \nu = 4.01$.

4.3. Sample Speed vs. Quality Tradeoffs for PSLD

Sampler Setup: Since the tradeoff between sample quality and the number of reverse sampling steps required is crucial for any SGM backbone, we now examine this tradeoff for PSLD for the CIFAR-10 dataset (See Appendix D.2 for extended results on the CelebA-64 dataset). We use our VP-SDE and PSLD with $\Gamma = 0$ (corresponding to CLD) ablation models as comparison baselines. Furthermore, we use combinations of the EM and SSCS samplers with Uniform (US) and Quadratic (QS) timestep striding as different sampler settings to benchmark the performance of all methods. It is worth noting that the SSCS sampler can only be used for augmented SGMs like CLD and PSLD. For ODE-based comparisons, we use the probability flow ODE setup with RK45 [10] solver (see Appendix C.5 for more details). Lastly, we measure sample quality using FID computed for 10k samples (denoted as FID@10k).

Main Observations: Table 5 shows a comparison between FID scores for our best performing PSLD models (corresponding to $\Gamma = 0.01$ and $\Gamma = 0.02$) and our VP-SDE and CLD baselines from Section 4.2 for the CIFAR-10 dataset across $N \in \{50, 100, 250, 500, 1000\}$ steps. We primarily observe that *PSLD outperforms the VP-SDE and CLD baselines across all comparison points* with the most significant differences at lower NFE (Network Function Evaluations)

Method	$\log_{10}$ tol	FID@10k (↓)	Avg. NFE
CLD (Baseline)	-5	5.54	280
	-4	5.62	196
	-3	6.54	147
	-2	9.98	86
	-1	397.1	27
(Ours) PSLD ($\Gamma = 0.02$)	-5	**4.79**	228
	-4	4.84	158
	-3	5.09	111
	-2	16.11	69
	-1	418.77	27
VPSDE	-5	5.91	123

Table 6: PSLD exhibits better speed vs. sample quality tradeoffs over competing baselines on CIFAR-10 using a black-box ODE solver. $\log_{10}$ tol indicates the ODE sampler (RK45) tolerance. **Bold** indicates best result for that column.

Method	FID (Train)	FID (Test)
CLD	1.01	7.10
(Ours) PSLD ($\Gamma = 0.01$)	**0.85**	**6.93**

Table 7: PSLD outperforms CLD on Image inpainting for the AFHQv2 dataset. FID (lower is better) is computed on the full train and test sets.

values. For instance, PSLD ($\Gamma = 0.02$) achieves the best FID of 16.12 at NFE=50 compared to 17.72 and 21.31 by VP-SDE and CLD, respectively. Moreover, for most sampler settings across methods, SSCS performs better in the low NFE regime ($N \leq 250$), while EM performs better for a higher number of NFEs. A similar observation was made in [9]. Similarly, Quadratic striding works much better in the low NFE regime, while Uniform striding works better when using a higher number of NFEs ($N > 500$).

We next compare our best-performing ablation model (PSLD with $\Gamma = 0.02$) with our VP-SDE and CLD baselines using the probability flow ODE setup across multiple tolerance levels on the CIFAR-10 dataset. Table 6 compares FID scores (computed on 10k samples) for all methods. Like our SDE setup, *PSLD (Γ=0.02) outperforms both baselines in sample quality for similar NFE budgets*. Moreover, across the same solver tolerance level, PSLD requires fewer NFEs on average than its CLD counterpart while yielding better sample quality. Lastly, we found using black-box solvers to further improve sample quality compared to the SDE baseline at a tolerance level of 1e-5 for both our PSLD and CLD models for CIFAR-10 (FID@10k=4.97 for PSLD for ODE vs. FID@10k=4.84 for EM-QS (N=1000) with $\Gamma = 0.02$). This observation is consistent with the ODE comparison results presented in Section 4.1.

4.4. Conditional Generation with PSLD

Following prior work [7, 56], given some conditioning information $\mathbf{y}$, an unconditional pre-trained score network $\boldsymbol{s}_\theta(\mathbf{u}_t, t)$ can be used for sampling from the distribution $p(\mathbf{z}_t|\mathbf{y})$ in PSLD. More specifically,

$$\begin{aligned}\nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t|\mathbf{y}) &= \nabla_{\mathbf{z}_t} \log p(\mathbf{y}|\mathbf{z}_t) + \nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t), \\ &\approx \nabla_{\mathbf{z}_t} \log p(\mathbf{y}|\mathbf{z}_t) + \boldsymbol{s}_\theta(\mathbf{z}_t, t). \end{aligned} \tag{13}$$

We can then use the estimate of $\nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t|\mathbf{y})$ in Eqn. 13 to sample from the following SDE for conditional generation:

$$d\mathbf{z}_t = \left[\boldsymbol{f}(\mathbf{z}_t) - \boldsymbol{G}(t)\boldsymbol{G}(t)^T \nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t|\mathbf{y})\right] dt + \boldsymbol{G}(t) d\mathbf{w}_t. \tag{14}$$

Figure 3 illustrates class conditional samples for the CIFAR-10 and the AFHQv2 datasets obtained by training an additional *time-dependent classifier* $p(\mathbf{y}|\mathbf{z}_t)$ to compute $\nabla_{\mathbf{z}_t} \log p(\mathbf{y}|\mathbf{z}_t)$, followed by sampling from the SDE in Eqn. 14 (full implementation details in Appendix D.4). Similarly, we can perform data imputation by setting the conditioning signal $\mathbf{y} = \bar{\mathbf{z}}_0$ where $\bar{\mathbf{z}}_0$ is the observed part of the input data $\mathbf{z}_0$ (See Figure 3). For image inpainting, PSLD exhibits a better perceptual quality of inpainted samples over CLD on the AFHQv2 dataset (See Table 7). We include a complete derivation for inpainting and an analogous framework to [56] for solving inverse problems using PSLD with additional conditional synthesis results in Appendix D.4.

5. Related Work

Advances in Diffusion Models: Following the seminal work on diffusion (a.k.a score-based) models [15, 50, 53, 56], there has been much recent progress in advancing unconditional [7, 9, 18, 19, 36, 42, 46, 58] and conditional [4, 38, 42, 44, 56] diffusion models for a variety of downstream tasks like text-to-image synthesis [37, 40], image super-resolution [27, 44] and video generation [14, 17, 63, 65]. Our work is closely related to CLD [9], which is motivated by Langevin heat baths in statistical mechanics [26]. However, our method is not directly motivated by physical interpretation but rather directly constructed from our proposed drift parameterization. Another line of research in SGMs is to perform score-based modeling in the latent space [42, 49, 58] of a powerful autoencoder [12, 57]. Such approaches have been shown to improve the sampling time in SGMs. Therefore, since we propose a novel diffusion model backbone, most existing advances in diffusion models complement PSLD.

Sampler Design in Diffusion Models: Improving the speed-vs-quality tradeoff in SGMs is a fundamental area in diffusion model research [2, 24, 29, 51, 66, 67]. One popular approach to speed up diffusion model sampling is DDIM [51]. [66] show that DDIM can be cast as an exponential integrator and propose further improvements. [67] further leverage these improvements to propose a



Figure 3: (Left) Class-conditional results on CIFAR-10: *Truck*, *Airplane*, and *Automobile* from Top to Bottom (two rows each). (Middle) Class conditional results on AFHQv2: *Dogs*, *Cats* and *Others* from Top to Bottom (one row each). (Right) Inpainting results on AFHQv2. The columns represent the original, corrupted, and imputed samples, respectively, from left to right.

generalized-DDIM (gDDIM) method for CLD. It is worth noting that gDDIM parameterization requires predicting the score w.r.t both the data and the auxiliary variables and is directly compatible with PSLD. Another line of research involves training to speed up diffusion sampling. GENIE [8] proposes to utilize higher-order Taylor methods during training to speed up DDIM sampling. Alternatively, distillation-based approaches distill a teacher into a student diffusion model progressively [34, 46] or otherwise [31]. Therefore, exploring some of these directions in the context of PSLD would be interesting.

Auxiliary Diffusion Models: In a concurrent work, [48] define an ELBO for multivariate diffusion models (MDM) and introduce a similar recipe as ours to design new diffusion processes. While [48] optimize for likelihood estimates, we primarily focus on sample quality in this work. Both works illustrate a different perspective on the advantages of constructing a generic recipe for designing diffusion processes and, therefore, complementary. Another recent work, Flexible Diffusion [11], exploits the geometry of the data manifold to parameterize the forward process. The proposed framework is complete under linear drift. However, our proposed parameterization makes no such assumptions.

6. Conclusion

We presented a recipe for constructing forward process parameterization for diffusion processes that guarantees convergence to a prespecified stationary distribution, such as a Gaussian. We use the proposed recipe to construct a novel diffusion process: Phase Space Langevin Diffusion(PSLD) which achieves excellent sample quality with better speed-vs-quality tradeoffs compared to existing baselines like the VP-SDE and CLD on standard image-synthesis benchmarks. We left the exploration of potentially performance-improving design choices such as alternative score network parameterizations and loss weighting [20] as directions for future work.

While this work only explores stochastic samplers with a single auxiliary "momentum" variable $\mathbf{m}$ (of the same dimension as $\mathbf{x}$), exploring other design choices of $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$ [48], which lead to higher-order stochastic samplers (like the Nosé-Hoover Thermostat) could also be an interesting research direction. Furthermore, our current choices of $\boldsymbol{D}(\mathbf{z})$ and $\boldsymbol{Q}(\mathbf{z})$ are limited to constant matrices due to relying on denoising score matching. Therefore, the proposed parameterization offers a complementary framework for designing diffusion generative models trained using alternative score-matching techniques.

Lastly, our proposed recipe is only *complete* under the assumption that both $\mathbf{x}$ and $\mathbf{m}$ are required to converge to prescribed marginals $p(\mathbf{x})$ and $p(\mathbf{m})$. Without this requirement on $\mathbf{m}$, the design space of samplers is potentially larger (as has been pointed out in the Bayesian MCMC literature) and may, e.g., include microcanonical samplers [41, 59]. However, the requirements of generative diffusion models are more strict and demand that the forward process's asymptotic joint distribution over $\mathbf{x}$ and $\mathbf{m}$ has a simple form that enables sampling in constant time. In contrast, Bayesian MCMC only requires the $\mathbf{x}$-marginal to converge to the prescribed posterior. We still think that relaxing the requirements on tractability enables potentially promising new samplers for future exploration.

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